PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

COAST REGION

REGIONAL FORM FOUR SECONDARY EDUCATION MOCK EXAMINATION

031/1

PHYSICS 1

(For Both School and Private Candidates)

Time: 3:00 Hours **Year: 2026**

Instructions

1. This paper consists of section A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **Sixteen (16) marks**, section B carries **Fifty – four (54)** marks and section C carries **Thirty (30) marks.**
4. Non –programmable calculators may be used.
5. Cellular phones and any unauthorized materials are **NOT** allowed in the examination room.
6. Write your Examination Number on every page of your answer sheet (s).
7. Where necessary the following constants may be used:
   (i) Acceleration due to gravity, $g = 10$ m/s.
   (ii) Speed of light in vacuum $= 3 \times 10^8$ m/s.
   (iii) Speed of sound in air = 340 m/s
   (iv) Specific heat capacity of water = 4200 J/kg^{0}C.
   (v) Specific heat capacity of aluminium = 900J/kg^{0}C.

SECTION A (16 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet provided.

i. What type of fire extinguisher should be used for a fire involving flammable gases?

A. Water extinguisher
B. Foam extinguisher
C. Carbon dioxide extinguisher
D. Dry powder extinguisher
E. Wet chemical extinguisher.

ii. Which of the following best explains why objects submerged in water appear to be closer to the surface than they actually are?

A. Light travels faster in water than in air, causing the object to appear closer.
B. Light rays from the object undergo total internal reflection before reaching the observer.
C. Light rays bend towards the normal as they pass from water to air, making the object appear raised.
D. The refractive index of water is lower than that of air, causing the light to diverge.
E. Light is scattered more in water, resulting in magnification of the object.

iii. What is the image distance and nature of image formed when an object is placed 15 cm in front of a convex mirror with the focal length of 10 cm?

A. 6cm, virtual, upright
B. 25cm, real, inverted
C. 6cm, real inverted
D. 15cm, virtual, upright
E. 25cm, virtual, upright.

iv. What type of fire extinguisher should be used for a fire involving live electrical equipment?

A. Water extinguisher
B. Foam extinguisher
C. Carbon dioxide extinguisher
D. Dry powder extinguisher
E. Wet chemical extinguisher.

v. Suppose an engine raises 400 kg of water steadily through a height of 40 m in 20 seconds. The upward force used is equal to the weight of water raised. Calculate the power in Kw.

A. 6 kW B. 7 kW C. 5 kW
D. 4 kW E. 8 kW.

vi. Which of the following media allows sound to travel fastest?

A. Steel B. Vacuum C. Carbon dioxide
D. Water E. Air.

vii. What factors does the gravitational force between two objects depends on?

A. The charge on the objects and the distance between them
B. The mass of the objects and the distance between them
C. The speed of the objects and their gravitational constant
D. The volume of the objects and their mutual acceleration
E. The weight of the objects and the direction of their motion.

viii. A certain substance is radioactive. If $15/16^{th}$ of the radioactive atoms of the substance decay in 8 minutes, determine the half-life of the substance;

A. 32 min B. 16 min C. 4 min

D. 2 min E. 1 min.

ix. Which of Newton's laws explains the recoil of a gun?

A. First law B. Second law C. Third law

D. Law of inertia E. Law of gravitation.

x. Which one of the following is not a property of cathode rays?

A. They are a stream of fast-moving protons.

B. They travel in straight lines.

C. They are deflected by electric fields.

D. They are deflected by magnetic fields.

E. They produce x – rays when stopped suddenly.

2. Match each part of the electromagnetic spectrum in list A with the correct function or use from list B then write the letter of the correct answer beside the item number.

List A	List B
i. Presbyopia	A. Eye focuses on object at different distance by varying the focal length of the lens
ii. Hyperopia	B. Gradual loss of the eye's ability to focus on near objects due to old age
iii. Myopia	C. Close the object appear clearly, but far ones do not
iv. Astigmatism	D. Imperfection in the curvature of the eye that cause the formation of blurred images.
v. CCTV Camera	E. Allow us to see object that are far away.
vi. Accommodation	F. Surveillance in high security installation, banks, shopping malls, offices and residences
	G. Refraction or reflection of light to form different types of image
	H. Image of the nearly object are blurry.

SECTION B (54 Marks)

Answer **all** questions from this section.

3. (a) In four ways describe how simple microscope differs from compound microscope. **(4 marks)**

(b) A converging lens forms an upright image that is four times the size of the object. If the focal length of the lens is 20 cm, determine the object distance. **(5marks)**

4. (a) With two reasons explain why aero planes are made up of aluminum alloys and not steel? **(3 marks)**

(b) A half metre rule AB is freely pivoted at 18 cm from end A and balances horizontally when a body of 35 g is hung 48 cm from end B. Calculate the mass of the rule. **(6 marks)**

5. (a) Explain four advantages of ocean tides. **(4 marks)**

(b) A 120N load is pushed up a straight smooth inclined plane. The angle between the inclined plane and the horizontal is 30^0. The load moved a distance of 5 m along the inclined plane. If the machine has an efficiency of 75%, calculate the effort required. **(5 marks)**

6. (a) Demonstrate by using diagram on how heat transfer by convection takes place when water is heated? **(3 marks)**

(b) A block of aluminium of mass 0.5 kg at a temperature of 100^0C is dropped in 1.0 kg of water at 20^0C. Assuming that no thermal energy is lost to the environment, what will final temperature of water be at thermal equilibrium? **(6 marks)**

7. (a) With the aid of diagram, compare the penetrating power of the three types of radiations on a piece of paper, aluminium sheet and lead block. **(4 marks)**

(b) Two sound waves produced beat frequency of 4 Hz. If one of the sound waves has a frequency of 20 Hz. Calculate the possible values of the frequencies of the other sound. **(5 marks)**

8. (a) Explain three ways that countries should adhere to minimize global warming in the world. **(3 marks)**

(b) A particular radioactive has a half-life of 2.0 hours. A sample gives count rate of 2400 per second at 11:00 am. When will the count have dropped to approximately 300 per second in the same counting system? **(6 marks)**

SECTION C (30 Marks)

Answer any two **(2)** questions from this section.

9. (a) A P-N junction is a semi-conductor material made by combining a P-type and N-type semiconductors in a single continuous crystal. Describe how the P-N junction operates. **(6 marks)**

(b) Semiconductor materials are very useful in many electronic components. By using your knowledge basing on a semiconductor, explain three properties of semiconductor materials. **(6 marks)**

(c) Mention three uses of X – rays. **(3 marks)**

10. (a) With the aid of diagram, explain how ammeter and voltmeter are connected in the electric circuit. **(4marks)**

(b) In certain circuit a current of 3 A of flows when the potential difference (p.d) is 2 V. When the current is reduced to 2V the potential difference (p.d) rise to 3V. Determine the e.m.f and internal resistance of the battery **(6 marks)**

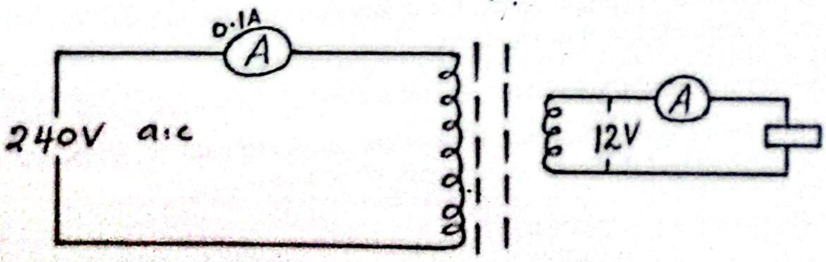
(c) A pipe closed at one end has a length of 100 mm. If the velocity of sound in air of the pipe is 340 m/s. Calculate the frequency of the first overtone. **(5 marks)**

11. (a) (i) Discuss the factors that determine the magnitude of the induced E.m.f in closed loop of wire. Give four points. **(2 marks)**

(ii) Briefly explain why a galvanometer connected to coil deflected when a magnet is moved toward the coil. **(4 marks)**

(b) State the application of the Induction coil. Use two points. **(2marks)**

(c) (i) The figure below shows a transformer used to stepdown power. Assuming that there is no power lose, what will be the ammeter reading on the output part? **(4 marks)**



(ii)An engineer determines that the power input to transformer is twice the power out. What would be the efficiency of the given transformer? **(3 marks)**